PORTFOLIO · SELECTED WORK 2024–2026

Davin Jonathan Tanus

AI Engineer, Information Systems, ITS

I build AI products end to end and measure them: a RAG assistant that cites the exact article of the ITS academic regulation, an in-browser sign-language recognizer, and evaluated NLP and vision models. At work I build LLM-based CV screening and document-intelligence tools. I care about honest evaluation and about shipping things people can use.

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Inside: Profile & education · Featured AI and machine-learning projects with outcomes · Leadership & achievements · Technical skills · Selected certifications

PROFILE

Information Systems undergraduate at Institut Teknologi Sepuluh Nopember (GPA 3.64/4.00) and AI Engineer Intern at PT Salam Pacific Indonesia Lines. My work spans **retrieval-augmented generation and LLM evaluation**, **machine learning** from data audit to deployment, and the full-stack engineering around them. I work in Agile teams, evaluate rigorously, and care about shipping products people actually use.

EDUCATION

Institut Teknologi Sepuluh Nopember (ITS)

Bachelor's Degree in Information Systems · Aug 2023 – Aug 2027 (expected) · GPA 3.64/4.00

- Thesis (in progress): bacterial vs. viral pneumonia classification on pediatric chest X-rays with RDNet-Tiny, compared against DenseNet121 and ConvNeXt-Tiny.
- Relevant areas: software engineering, data analytics, machine learning, IS/IT strategy, IT risk and service management.
- 2nd Place, Information Systems Programming Competition (HMSI ITS, 2024).

AI Engineer Intern, PT Salam Pacific Indonesia Lines

Performance Excellence Division · Applied AI & Full-Stack Engineering · Jul 2026 – Nov 2026

- Largest contributor to a fleet fuel (bunker) dashboard: analysis of vessel noon reports by route, speed, engine, and reefer load, with data checks for missing or inconsistent reports.
- Rebuilt the LLM-based CV analysis module of a talent platform into two recruiter workflows (one CV against all roles; many CVs ranked for one role) and built a DISC assessment analytics dashboard.
- Built a document-verification engine for fleet route plans, integrated apps with the company SSO, and added account management to the SSO portal (Next.js, FastAPI, Go, Laravel, PostgreSQL/MySQL, Docker).

TECHNICAL SKILLS

AI & Machine Learning: RAG with hybrid search, LLM evaluation, ONNX Runtime, TensorFlow, PyTorch, scikit-learn, transfer learning, BiLSTM, time-series forecasting, on-device deployment (TensorFlow Lite, TensorFlow.js, SavedModel)

Software & Data: JavaScript, HTML, CSS, Python, SQL, Go · React, Next.js, FastAPI, Streamlit, REST APIs · PostgreSQL, MySQL, MongoDB

Cloud, DevOps & Ways of Working: Power BI, Microsoft Fabric, Excel, PowerPoint · Docker, GCP + Cloud Run, GitHub Actions CI/CD, Git · Agile Scrum/Kanban, product-oriented delivery

LEADERSHIP & ACHIEVEMENTS

Scrum Master, Agile Snacklicious

Scrum delivery · Six-person team · Four sprints

- Led sprint execution, daily scrums, and retrospectives, divided tasks, and tracked income and expenses.

Intern Staff, HMSI ITS (Information Systems Student Association)

Sep 2024 – Dec 2024

2nd Place, IS Programming Competition

HMSI ITS · 2024

FEATURED PROJECTS: AI PRODUCTS

RAG Assistant for the ITS Academic Regulation Live demo | GitHub

Role: AI engineer · personal project · 57-page official regulation, 91 articles

- **Problem:** students need exact answers from a long academic regulation, and an AI answer is only useful if it points to the article it came from.
- **Approach:** OCR-tolerant chunking per Pasal and ayat; BM25 and multilingual-e5 embeddings fused by reciprocal rank fusion; the LLM must cite retrieved passages and the server drops any other citation; quantised ONNX runtime without PyTorch (about 285 MB of RAM); FastAPI, Docker, pytest, and CI that fails if retrieval drops.
- **Outcome:** on a frozen 70-question set, retrieval MRR 0.84 against 0.80 for keywords alone, and all 10 questions outside the regulation refused; a hand check of the LLM judge is published with the failure cases.

RAG | Hybrid Search | LLM Evaluation | FastAPI | ONNX

BISINDO Fingerspelling, In-Browser Sign Language Recognition Live demo | GitHub

Role: ML engineer & front-end · personal project · 26 letters, on-device

- **Problem:** read the Indonesian Sign Language alphabet from a webcam without a server, and know how far the accuracy can be trusted.
- **Approach:** audited the public dataset (63% of validation images were near-copies of training images) and re-split by perceptual-hash groups; fine-tuned YOLO11n, chose on validation only, exported to ONNX and ran it client-side with ONNX Runtime Web plus an 8-frame vote.
- **Outcome:** 98.5% on a held-out test set (scored once) and 93.6% on an unseen signer; every error there falls on O or Y, and the failure analysis is published on the live demo page.

YOLO11 | PyTorch | ONNX Runtime Web | Data Leakage Audit

LLM Extraction from SEC 10-K Filings, Scored Against XBRL GitHub

Role: AI engineer · personal project · 39 filings, 220 values

- **Problem:** LLMs are often used to pull numbers out of financial reports, but they are rarely checked against a trusted answer.
- **Approach:** BM25 retrieval picks the statement pages; the LLM returns each figure as printed, its unit, and a quote; the code converts units and checks that the quote is in the filing. The answer key is the company's own XBRL fact for that exact filing.
- **Outcome:** 211 of 220 values correct (95.9%) with 16 excerpts, 84.5% with 8, and no thousand or million mix-ups. Four of the nine misses were revenue lines that mean something different from the XBRL figure.

LLM | Information Extraction | BM25 | SEC EDGAR | XBRL

FEATURED PROJECTS: MACHINE LEARNING

Indonesian App Review Sentiment Analysis GitHub

Role: ML engineer · rebuilt from a Dicoding project · 16,000 self-collected reviews

- **Problem:** tell positive from negative Indonesian app reviews, and know how far the score can be trusted.
- **Approach:** labels from star ratings only, slang normalisation, de-duplication; TF-IDF word + character features with LinearSVC against logistic regression and a majority baseline; per-class metrics and error analysis.
- **Outcome:** 91.8% accuracy (macro-F1 0.918) on 2,009 held-out reviews. The original 92% three-class figure came from labels built partly from the text and from dropping hard cases; honest three-class macro-F1 is 0.60.

NLP | scikit-learn | Evaluation | Label Quality

Stock Forecasting: Honest Walk-Forward Test (UNTR.JK) GitHub

Role: ML engineer · personal project · 250 walk-forward test days

- **Problem:** can ARIMA, XGBoost, or an LSTM forecast tomorrow's close better than "tomorrow equals today"?
- **Approach:** one-step walk-forward over a year where each forecast sees only earlier prices (asserted by a test), refits on a schedule, MAPE, direction accuracy, and Diebold-Mariano tests against the naive forecast.
- **Outcome:** no model beat naive (1.83% MAPE); direction 44–47%. This replaced an earlier result of mine (1.04%) that came from a one-day misalignment.

Time Series | Walk-Forward | ARIMA | XGBoost | LSTM

Multi-Class Image Classification with On-Device Export Training notebook | Case study

Role: ML engineer · Dicoding deep-learning project

- **Problem:** accurate image recognition that can run beyond the lab: server, mobile, and browser.
- **Approach:** MobileNetV2 transfer learning on 13,394 Imagenette images (10 classes, 80/10/10 split), augmentation, staged fine-tuning, checkpointing; evaluated with learning curves, confusion matrix, per-class metrics.
- **Outcome:** 91.67% test accuracy; model exported to SavedModel, TensorFlow Lite (mobile), and TensorFlow.js (browser).

TensorFlow | MobileNetV2 | Transfer Learning | TFLite | TensorFlow.js

SELECTED CERTIFICATIONS

Belajar Fundamental Deep Learning, Dicoding Indonesia (Sep 2026)

Computer vision, NLP, evaluation, deployment workflows

Cisco DevNet Associate, Cisco Networking Academy × ITS (Jul 2026)

APIs, automation, application security, testing, cloud deployment

AWS AI Academy, Basic & Beginner, Dicoding × AWS (Aug 2026)

AI foundations on cloud infrastructure

LangChain Academy Foundations, LangChain, LangGraph, LangSmith (Aug 2026)

LLM applications, agentic workflows, deployment operations

Full archive (24 certifications + learning activities with verification documents): **davintanus.vercel.app/credentials**

MORE SELECTED WORK

Conduit CI/CD on Google Cloud (Docker, GitHub Actions, Cloud Run), Academic Data Warehouse & ETL, BCA KPI dashboards, Banana-Quality clustering (K-Means + PCA), ISO 31000 risk management, IS/IT strategic planning, and computer-vision simulation, all documented with reports and previews at **davintanus.vercel.app/project**.

> **Let's build something people love to use.**
> I learn fast, collaborate in Agile teams, and finish what I start, from model notebook to deployed product. Reach me at **davinjtanus@gmail.com** or **LinkedIn**.